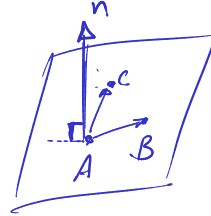


Week 8

1. Write equation of plane containing $A(3 \ 0 \ 1)$, $B(1 \ 1 \ 0)$, and $C(4 \ -2 \ 0)$,

$$\vec{AB} = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}, \vec{AC} = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$$



$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{bmatrix} -3 \\ 3 \\ 3 \end{bmatrix}$$

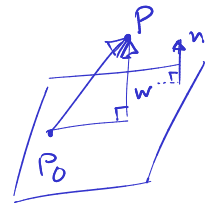
An equation of the plane: $\vec{n} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{n} \cdot \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

$$-3x + 3y + 3z = -3 \cdot 3 + 3 \cdot 0 + 3 \cdot 1$$

$$\boxed{-3x + 3y + 3z = -6}$$

2. Find the shortest distance from $P(1, 2, 3)$ to the plane $x - z = 2$ $\leftarrow n = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$
 $\hookrightarrow \|w\|$

$w = \text{Proj}_n P_0P$, where P_0 is any point on the plane.



Let $P_0(2, 0, 0)$, then $P_0P = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$

$$w = \frac{P_0P \cdot n}{n \cdot n} n = \frac{-1-3}{1+0+1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}$$

$$\|w\| = \sqrt{4+0+4} = \sqrt{8} \leftarrow \text{distance}$$

3. Find shortest distance between 2 skew lines

$$L_1 = \begin{bmatrix} 1 & -1 & 0 \end{bmatrix} + t \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

$$L_2 = \begin{bmatrix} 2 & -1 & 3 \end{bmatrix} + t \begin{bmatrix} 3 & 1 & 0 \end{bmatrix}$$

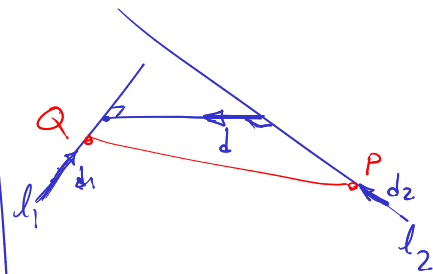
$W = \text{Proj}_d PQ$, where P is any point on l_2 and Q is any point on l_1 . Let

$$Q(1, -1, 0), P(2, -1, 3).$$

$$\text{Then } \vec{PQ} = \begin{bmatrix} -1 \\ 0 \\ -3 \end{bmatrix}.$$

$$W = \frac{PQ \cdot d}{d \cdot d} d = \frac{1+0+6}{1+9+4} \begin{bmatrix} -1 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -3/2 \\ -1 \end{bmatrix}$$

$$\|W\| = \sqrt{\frac{1}{4} + \frac{9}{4} + 1} = \frac{\sqrt{14}}{2} \leftarrow \text{distance}$$

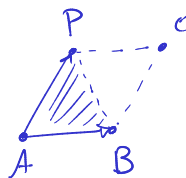


$$d = d_1 \times d_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ -2 \end{bmatrix}$$

4. Find the area of triangle determined by A (3, 1, 0), B (1, 1, 0) and P (4, -2, 0)

$$AB = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, AP = \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}$$

$$AB \times AP = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}$$



$$\text{Area of } \triangle ABP = \frac{\text{Area of } \square ABP}{2}$$

$$= \frac{\|AB \times AP\|}{2}$$

$$= \frac{6}{2} = 3$$

↑
area of $\triangle ABP$

5. The volume of parallelepiped determined by u , v , and w is $|u \cdot (v \times w)|$.

Find the volume of the parallelepiped determined by $u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

$$|u \cdot (v \times w)| = \left| \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) \right| = \left| \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -6 \\ 4 \end{bmatrix} \right| = |0 - 6 + 4| = 2$$

OR

$$|u \cdot (v \times w)| = \left| \begin{vmatrix} 1 & 1 & 1 \\ 1 & -2 & -3 \\ 1 & 2 & 3 \end{vmatrix} \right| = |-6 - 3 + 2 + 6 - 3 + 2| = |-2| = 2 \quad \text{Volume.}$$

6. If T is a linear transformation and

$T[1, -1, 2] = [3, -2, 1, 0]$ $T[1, 0, -1] = [1, 1, 2, -2]$, then what is

$T[-1, -1, 4]$ equals to?

$$c_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix} \xrightarrow{\text{solve}} \left[\begin{array}{cc|c} 1 & 1 & -1 \\ -1 & 0 & -1 \\ 2 & -1 & 4 \end{array} \right] \rightsquigarrow \begin{cases} c_1 = 1 \\ c_2 = -2 \end{cases}$$

$$\begin{aligned} T \left(\begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix} \right) &= T \left(\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right) = T \left(\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \right) - 2 T \left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right) \\ &= \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -3 \\ 4 \end{bmatrix}. \end{aligned}$$